8.5 Greyhounds Chasing Greyhounds

WORKSHEET

The question

STEP III 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself By symmetry, what shape do the four dogs form at every moment? Where is its centre?

STAGE 1 The direction of motion

Ask yourself A runs towards B. In the triangle OAB, what angle does AB make with AO (the direction towards the centre)?

STAGE 2 The differential equation

Ask yourself Split A's velocity into radial and transverse parts. In polar coordinates the radial velocity is $\dot{r}$ and the transverse velocity is $r\dot{\theta}$.

STAGE 3 n dogs

Pause and try For a regular n-gon, what is the angle at O in triangle OAB? What angle does AB make with AO? Repeat Stage 2.

What's really going on

Infinitely many turns, finite distance

The radial speed $\dot{r} = -v\sin\frac{\pi}{n}$ is constant, so the dogs reach the centre after time $T = \frac{r_0}{v\sin(\pi/n)}$, having each run a distance $vT = \frac{r_0}{\sin(\pi/n)}$. For a regular n-gon of side s, $r_0 = \frac{s}{2\sin(\pi/n)}$, so each dog runs

$$\frac{s}{2\sin^2(\pi/n)}.$$

For four dogs this is exactly s: each dog runs precisely the length of one side. (Here is a slick way to see it: B's velocity is always perpendicular to AB, so it does not change the distance AB; only A's motion, at speed v straight along AB, closes the gap, and the gap is initially s.) For three dogs the distance is $\frac{2}{3}s$, and for six it is $2s$. Yet in every case the spiral winds round the centre infinitely many times, because $\theta \to \infty$ as $r \to 0$. A curve of finite length with infinitely many turns is no paradox: the turns get smaller fast enough.

The logarithmic spiral

The curve $r = \lambda e^{-k\theta}$ is the *logarithmic* or *equiangular* spiral: it crosses every ray from the centre at the same angle. It appears in nature, in the shell of the nautilus and the arms of some galaxies, because growth that preserves shape produces it. It fascinated Jacob Bernoulli so much that he asked for it to be carved on his tombstone with the motto *Eadem mutata resurgo* ("Changed, I shall arise the same"), referring to the fact that it looks the same at every scale. The stonemason carved an Archimedean spiral instead.

Pursuit in practice

"Pure pursuit", always heading straight at the target, is the simplest guidance law for a missile or a predator. It is not the best: heading for where the target *will* be

("proportional navigation") is more efficient, and is what both modern missiles and some predators, such as dragonflies and falcons, actually do. The mathematics of pursuit and evasion became a subject of its own in the twentieth century under the name *differential games*.

Try this next

1. **The slick argument.** Explain carefully why the distance AB decreases at rate exactly v in the four-dog problem, and deduce that the dogs meet after time s/v.

2. **Three dogs.** Three dogs start at the corners of an equilateral triangle of side s. Find the equation of each path, the time until they meet, and the distance each runs.

3. **How many turns?** For four dogs starting 100 m from the centre, how close to the centre are they after one full turn? After how many full turns are they within 1 mm?

4. **Two dogs.** What happens with $n = 2$ (two dogs facing each other)? Does the formula still make sense?

Further reading

- **Paul Nahin, *Chases and Escapes: The Mathematics of Pursuit and Evasion*** (Princeton University Press). Everything about pursuit curves, from Bouguer to missiles.
- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). Includes the logarithmic spiral and Bernoulli's tombstone.
- **D'Arcy Thompson, *On Growth and Form*** (Cambridge University Press). The classic book on mathematics in nature, including spiral growth.

The full guided solution is in Section 8.5:
https://beyondstep.keithmcnulty.org/ch08/pursuit.html